

TPC GROUP 14

Programming Life Cycle

Analysis through evaluation

Client: SkillsTrack Training Centre

Project: Learner Support Portal

Programme: Technical Programming Cycle | September 2026

Programming Life Cycle

Requirement Analysis (What do we need?)

- Figure out the problem you're solving.
- Talk to the people who will use the program to understand their needs.
- Set clear goals and limits.

Design (How will it work?)

- Sketch out the big picture of the system.
- Plan the details: what parts (modules) you'll build, what data you'll use, and what steps (algorithms) the program will follow.
- Use diagrams or flowcharts to make it clear.
- Example: Draw a simple diagram showing learners logging in, adding tasks, and assessors viewing progress.

Implementation (Coding - Build it)

- Write the actual code based on your design.
- Follow good coding habits so it's clean and easy to maintain.
- Use tools like Git to keep track of changes.
- Example: Build a "Book Support" button that saves the booking in Firebase.

Testing (Does it work?)

- Check small pieces of code (unit testing).
- Make sure all parts work together (integration testing).
- Test the whole system against the original requirements.
- Let real users try it (user acceptance testing).

- Example: Test if a learner's progress updates correctly when they finish a task.

Deployment (Put it out there)

- Move the program from development to real use.
- Set up servers, databases, and security.
- Give users instructions or training.
- Example: Launch at skills-track.firebaseio.com.

Maintenance (Keep it running)

- Fix problems that show up after launch.
- Update the program when needed.
- Add new features if users ask for them.
- Example: Update the portal when Firebase changes its rules.

Evaluation & Retirement (Is it still useful?)

- Check if the program is still doing its job well.
- Decide if it should be improved, replaced, or shut down.
- Example: Upgrade from Firebase to a full-stack system if learner numbers grow a lot.